

Shyamal Dharia

[\[GitHub\]](#) [\[Google Scholar\]](#) [\[LinkedIn\]](#) [\[Email\]](#) [\[Academic Website\]](#)

EDUCATION

University of Winnipeg

Masters in Applied Computer Science

Winnipeg, Canada

2022 – 2024

- GPA: 4.4/4.5—Graduated with Highest Distinction

Gujarat Technological University

Bachelor of Engineering in Electronics & Communication

Surat, India

2017 – 2021

- GPA: 7.9/10

PUBLICATIONS

Best Paper Award:

- ***Differentiable Logic Gate Networks for Low-Latency EEG Classification on Edge Devices***
S. Y. Dharia; S. D. Smith; C. E. Valderrama (Oral Presentation at CAI 26; 20%)
(Canadian AI Conference 2026 – **Best Paper Award**)

Pre-prints or Under Review:

- ***Zero-Shot Harmonization of Multi-Site Resting-State fMRI with Joint-Embedding Predictive Architectures***
S. Y. Dharia; S. D. Smith; C. E. Valderrama
Under Review (Journal)
- ***Cross-Cohort Transfer of Clinical Alzheimer's EEG Signatures to Pre-Clinical APOE-ε4 and PICALM rs3851179 Risk Carriers Using Differentiable Logic Gate Networks***
S. Y. Dharia; C. E. Valderrama; S. D. Smith
Under Review (Journal)

Journal Publications:

- ***Exploring the Interaction of APOE-ε4 and PICALM rs3851179 With Dynamic Functional Connectivity in Healthy Middle-Aged Adults at Risk for Alzheimer's Disease***
S. Y. Dharia; C. E. Valderrama; Q. Liu; S. D. Smith
IOP Journal of Neural Engineering, 2026
<https://doi.org/10.1088/1741-2552/ae4926>
- ***A Novel Approach for the Early Identification of Genetic Risk Factors for Alzheimer's Disease Using EEG and Psychometric Data***
S. Y. Dharia; Q. Liu; S. D. Smith; C. E. Valderrama
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3639217>
- ***Fractal Dimension of Resting-State EEG as a Biomarker for Autonomous Sensory Meridian Response (ASMR) — (Oral Presentation at BHI 25; 9.2%)***
S. Y. Dharia; C. E. Valderrama; Q. Liu; B. K. Fredborg; A. S. Desroches; S. D. Smith
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3612301>
- ***Dual-Transformer Cross-Attention Framework for Alzheimer's Disease Detection via dPTE-Guided EEG Channel Selection and Multi-Modal Integration***
S. Y. Dharia; Q. Liu; S. D. Smith; C. E. Valderrama
Biomedical Signal Processing and Control, 2025
<https://doi.org/10.1016/j.bspc.2025.108390>
- ***Bipartite Graph Adversarial Network for Subject-Independent Emotion Recognition***
M. Niaki; S. Y. Dharia; Y. Chen; C. E. Valderrama
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3570187>

Conference Publications:

- ***Using Machine Learning to Model EEG-Derived Brain Activity During Emotion Regulation***

M. Hojjati; **S. Y. Dharia**; S. G. Camorlinga; S. D. Smith; A. S. Desroches; B. D. Brenneman
Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Copenhagen, Denmark, 2024
<https://doi.org/10.1109/EMBC58623.2025.11253722>

- **CNN-Based Chagas Disease Detection with 12-lead ECG**
S. Y. Dharia; M. Hojjati; S. Rahman; M. M. T. Nur; C. E. Valderrama
Computing in Cardiology Conference (CinC) 2025 — PhysioNet Challenge
[\[Link\]](#)
- **Dataset-Independent EEG Channel Selection for Emotion Recognition**
S. Y. Dharia; S. G. Camorlinga; C. E. Valderrama; M. Hojjati
Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Orlando, FL, USA, 2024
<https://doi.org/10.1109/EMBC53108.2024.10782444>
- **Leveraging Machine Learning and Threshold-Free Cluster Enhancement to Unravel Perception of Emotion and Implied Movement**
S. Y. Dharia; M. Hojjati; S. G. Camorlinga; S. D. Smith; A. S. Desroches
IEEE EMBS International Conference on Biomedical and Health Informatics (BHI), Houston, TX, USA, 2024
<https://doi.org/10.1109/BHI62660.2024.10913664>
- **Multimodal Deep Learning Model for Subject-Independent EEG-Based Emotion Recognition**
S. Y. Dharia; C. E. Valderrama; S. G. Camorlinga
IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), Regina, SK, Canada, 2023
<https://doi.org/10.1109/CCECE58730.2023.10289007>

RESEARCH EXPERIENCE

University of Winnipeg	Winnipeg, Canada
<i>Senior Research Assistant</i>	2025 – Present
• Conduct machine learning and statistical analysis on multimodal neuroimaging data (EEG, fMRI, sMRI) to study biomarkers of Alzheimer's Disease, ASMR, and other neurological conditions. • Published peer-reviewed papers in leading journals/conferences, contributing to the fields of biomedical signal processing and computational neuroscience.	

Misericordia Health Center	Winnipeg, Canada
<i>Mitacs Accelerate Intern</i>	May 2024 – Dec 2024
• Developed an EEG data collection pipeline using PsychoPy with custom experiment integration, enabling two successful EEG-based emotion regulation studies with 60+ participants using wireless OpenBCI devices. • Designed and trained machine learning models for emotion recognition and intensity prediction, leading to two conference publications.	

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, scikit-learn, NumPy, pandas) and C++ (CUDA)
Signal Processing & Neuroimaging: MNE-Python, Nilearn, SciPy, fMRIPrep
Software & Development: Linux/Windows, Git, Docker, ROS (Robot Operating System)

AWARDS

- 2025 - Graduate Student of Highest Distinction
- 2024 - EMBC NextGen Scholar

LEADERSHIP EXPERIENCE

University of Winnipeg	Winnipeg, Canada
<i>CinC Physionet 2025</i>	2025
• Led a team in the PhysioNet/Computing in Cardiology Challenge on Chagas disease detection from ECG signals with a CNN-based morphological features and lead-wise learning approach.	

TEACHING ASSISTANT & ACADEMIC SERVICES

- **Associate Editor**—EMBC 2026

- **Peer Reviewer**—International Journal of Human-Computer Interaction (2026)
- **Peer Reviewer**—Public Library of Science (PLOS) *Aging and Health* (2026)
- **Peer Reviewer**—IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 26)
- **Peer Reviewer**—IEEE EMBS International Conference on Biomedical and Health Informatics (BHI 26 and 25)
- **TA**—GACS-7203 Pattern Recognition (Fall 2024 and 2023—*Dr. Camilo E. Valderrama*)
- **TA**—GACS-7206 Advance Machine Learning (Winter 2024—*Dr. Sheela Rammanna*)

GRANT WRITING EXPERIENCE

Mitacs Accelerate (2023) | \$60,000 (Awarded)

- Co-drafted the initial research-industry proposal in collaboration with PIs for a successful partnership between the University of Winnipeg and Misericordia Health Center.

NSERC Alliance (2026) | \$30,000 (Awarded)

- Contributed to the drafting of a collaborative grant proposal involving University of Winnipeg and Interaxon.

UWinnipeg Major Research Grant (2026) | \$7,500 (Awarded)

- Assisted the PI in preparing the proposal to support direct research costs and laboratory operations.